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Malicious SMS detection using ensemble learning and SMOTE to improve mobile cybersecurity

Hongsheng Xu [a], Akeel Qadir [b], Saima Sadiq [c,*]

[a] Industry-Education-Research Institute of Advanced Materials and Technology for Integrated Circuits, Anhui University, 111 Jiulong Road, Hefei, Anhui 230601, P.R. China
[b] School of Information Engineering, Xi'an Eurasia University, 710065, China
[c] Khwaja Fareed University of Engineering & Information Technology, Pakistan

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ABSTRACT

The widespread use of cell phones, along with their constant internet connection, makes them vulnerable to malicious SMS attacks, including smishing and spam. Smishing involves attempts to steal personal information, while spam focuses on unwanted advertisements. Both pose cybersecurity threats, often requiring effective filtering techniques. Researchers have devised multiple methods for detecting malicious SMS, yet a notable gap remains in creating algorithms to reduce false positives, where normal messages are wrongly classified as malicious. The method employs ensemble learning to automatically identify malicious or legitimate messages. It combines Support Vector Machine and Random Forest models, compared with individual machine learning approaches for smishing detection. Feature extraction methods like Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF) are employed to derive features from the data. The imbalanced issue of the dataset is addressed by applying the Synthetic Minority Oversampling Technique (SMOTE). The results showed that the ensemble model outperformed the individual models, with an accuracy score of 99.58% when trained using TF–IDF on the balanced dataset. The proposed approach offers proactive defense against malicious SMS attacks, enhancing cybersecurity in the mobile communications sector.

1. Introduction

The advancement of information technology has led to widespread internet access and a growing dependence on smartphones. As these devices become more affordable and accessible, their use continues to rise globally, connecting people across diverse demographics (Bajpai and Radha, 2019). The spike in smartphone usage has accelerated the integration of technology into daily life, enhancing communication, education, and commerce (Madrid et al., 2022). However, the growing number of internet users has also fuelled a rise in cybercrimes, particularly phishing attacks (Ibrahim and Obaid, 2021). The storage of private information on smartphones, including credit cards and banking details, has made user privacy a critical issue in the world of smart gadgets. Adversaries aim to acquire confidential data such as user passwords and financial details through the act of masquerading as a reputable organization (Sonowal and Sonowal, 2022). Phishing attempts pose a particular risk to mobile phone users due to several factors, including challenges in verifying the legitimacy of requests, the adoption of risky user behaviours, and the readiness of mobile users to provide their credentials when asked (Desolda et al., 2021).

Smishing, a portmanteau of SMS (Short Message Service) and Phishing, was first coined by McAfee's David Rayhawk in 2006 (Sonowal and Sonowal, 2022). Smishing, or SMS-based phishing, involves the transmission of deceptive SMS messages to a target with the aim of coercing them into divulging personal information to the attacker, commonly referred to as the 'smisher'. Fig. 1 illustrates a sample smishing SMS, demonstrating how attackers attempt to deceive recipients through fraudulent messages. SMS serves as a widely used messaging platform, enabling users to transmit concise text-based messages to one another. Given its cost-effectiveness in reaching a broad audience, text messaging finds extensive application in advertising (Brown and Greenfield, 2021). The rising frequency of text message use has proven advantageous to malicious actors. Notably, more than 90% of SMS messages are opened in less than three seconds (Wahsheh and Al-Zahrani, 2022), indicating a sense of urgency that can be exploited in cyberattacks. In SMS phishing, the attacker sends messages to the recipient, typically containing links that direct them to malicious websites, applications, or user interfaces, where they are prompted to disclose their credentials. SMS is favoured over email for phishing attacks (Canova et al., 2015)

* Corresponding author.
E-mail address: s.kamrran@gmail.com (S. Sadiq).

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Fig. 1. Smishing SMS sample.

due to the rapid response rate to text messages, the ability to send SMS messages in bulk, and the constraints of small mobile device displays.

Among the various communication channels available, including phone calls, SMS, email, and more, text messaging is the preferred choice for attackers when it comes to conducting phishing attacks (Alabdan, 2020). Attackers prefer text messages due to the elevated responsiveness of recipients. Research conducted by CallHub has demonstrated that the response rate to SMS messages significantly surpasses that of email (Delany et al., 2012). Phishing and smishing attacks saw a significant increase in frequency during the COVID-19 outbreak. Smishers targeted COVID-19 patients throughout this worldwide crisis, claiming to have blood plasma and COVID-19-related materials that were reportedly obtained from government agencies operating remotely. The United Nations (UN) reported a 350% rise in phishing attacks during the pandemic (Gradoń, 2020).

Smishing has been studied in great detail, and several academics have shared different methods for spotting smishing communications. Numerous of these studies suggested fixes include explicit instructions, such as alerting users about strange URLs in SMS messages and similar things. Nevertheless, there has not been much discussion of the difficulties in identifying smishing indications in the literature. According to a survey of current methods, a sizable percentage of them rely on strategies for smishing attack detection such blacklisting, heuristics, and whitelisting (Ghourabi, 2021). Despite this, existing methods like blacklisting have limitations, as attackers regularly change URLs and employ techniques that avoid detection. Notably, because attackers regularly use URL shortcut services and smishing URLs change swiftly, the blacklisting strategy has shown to be limited. As a result, they often change domain names for phishing efforts, making blacklisting techniques useless. Some researchers have developed methods that look for certain words and URLs in text messages to identify smishing attacks. However, it is important to note that these indicators can also be found in legitimate messages from service providers and websites. Therefore, messages should not be labelled as smishing unless there is clear evidence that the URL leads to a malicious website trying to steal credentials or download malware. It is important to verify the contents of the URL before definitively labelling a message as smishing (Jain et al., 2020). Existing methods for detecting smishing messages have certain limitations. Hence, there is a need to formulate a new and efficient approach for identifying smishing messages while mitigating the occurrence of false-positive results. The approach can be deployed in real-world mobile environments, protecting users from financial loss and privacy breaches. Furthermore, this system can be integrated into mobile security apps or telecom networks, providing real-time protection for a growing population of smartphone users, particularly in regions where mobile phone penetration is high but digital literacy remains low.

Earlier research studies presented a framework to address the smishing challenge. The authors utilized a flow-based methodology to tackle the issue, taking into consideration various facets of the SMS content and the associated URL (Mishra and Soni, 2020). The previously mentioned approaches involved data classification through the utilization of machine learning algorithms. The objective of this study is to construct an efficient malicious SMS detection system by leveraging the capabilities of an ensemble learning model. This study offers the subsequent contributions:

- An ensemble learning method is proposed that combines Support Vector Machine (SVM) and Random Forest (RF) models to accurately identify smishing and unwanted SMS attempts.
- To prevent the proposed model from overfitting, the class imbalance issue in the dataset is addressed using the Synthetic Minority Oversampling Technique (SMOTE).
- To improve malicious SMS detection by exploring different text message feature extraction methods, this study employed Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF).
- To evaluate the performance of ensemble learning models compared to individual machine learning methods for distinguishing between malicious SMS and legitimate SMS.

The remaining segments of the paper are organized as follows: In Section 2, prior studies about the detection of smishing SMS are reviewed. Section 3 elaborates on the materials and methodologies used in the experiments to improve effectiveness. Section 4 presents the outcomes and ensuing discussion. Lastly, Section 5 finally concludes the study and provides an exploration of their implications.

2. Related work

In recent years, scholars have shown increased interest in smishing due to its prevalence in mobile smishing attacks. Several research endeavours have devised techniques for recognizing smishing messages (Mishra and Soni, 2019; Alani and Tawfik, 2022; Raminedi et al., 2023). Numerous researchers have explored smishing strategies and identification approaches to increase awareness among both users and researchers (Kamau et al., 2022; Ulfath et al., 2022; Saeki et al., 2022; Sharaff et al., 2023). Some studies also discuss certain methods for spam identification that have utilized comparable methodologies (Ghourabi et al., 2020; Kotsifakou et al., 2022). This section presents a compilation of research contributions from other scholars in the field of smishing detection. We have encompassed various techniques for detecting smishing as well as methods employed in the realm of phishing detection (Huang et al., 2019; Catal et al., 2022; Raminedi et al., 2023).

The authors proposed a smishing identification method referred to as the 'smishing classifier' to detect smishing messages (Goel and Jain, 2018). Utilizing the Machine learning Classifier, this system scrutinizes the SMS content and associated keywords. The suggested technique verifies the existence of URLs within the SMS and scrutinizes the sender's phone number. This model also assesses the appearance of the linked webpage and the initiation of file downloads. In another study (Sonowal and Kuppusamy, 2018), researchers introduced an approach known as SmiDCA to detect smishing messages through a supervised model. Employing a correlation-based methodology, the proposed model extracted thirty-nine attributes from smishing SMS. Supervised Machine Learning methods were applied, and assessments were conducted across diverse datasets.

A study introduced a rule-based approach for identifying smishing messages (Jain and Gupta, 2018). The research identified nine guiding principles for discriminating smishing content from legitimate messages. These principles were then imparted using various classification

techniques, including PRISM. Scholars utilized a content-driven method to detect smishing SMS messages (Mishra and Soni, 2019). A machine learning approach was employed to identify the prevalent terms in smishing SMS content. Furthermore, this model examines the legitimacy of the login page, the downloading of .apk files, and assesses the potential malicious nature of URLs. Researchers devised a feature-driven model, harnessing the power of machine learning techniques, to detect smishing SMS messages (Jain and Gupta, 2019). This model extracted a range of distinctive attributes from smishing messages and conducted an evaluation of these attributes using five distinct Machine Learning methods on the same dataset.

The authors (Mishra and Soni, 2019) outlined a variety of smishing tactics executed by assailants. Furthermore, the author proposed specific countermeasures that users can adopt to thwart smishing assaults. The research also delves into a range of strategies and techniques for combating smishing attacks. The authors (Catherine, 2019) introduced a method for categorizing fraudulent websites and scrutinized effective factors in the identification of deceptive sites. An algorithm based on rules was developed, leveraging the frequency of attributes in their datasets to establish the typical traits for identifying fraudulent websites. The author devised a model called 'PhiDMA' comprising five levels for detecting phishing websites, which included an interface designed for visually-impaired individuals (Sonowal and Kuppusamy, 2020). In Alswailem et al. (2019), a method for detecting phishing websites involves computing TF–IDF scores for individual words in documents and selecting the top five words to create a lexical signature, where a website is considered legitimate if its domain name matches one of the top N search results; otherwise, it is classified as a phishing website.

The authors in their study (Alabdan, 2020) illustrated diverse types of phishing attacks along with the technical methods employed. They also addressed several measures to mitigate phishing threats and advocated for best practices to enhance the security of mobile devices, ultimately aiming to heighten security awareness among mobile users. The authors presented a method for inspecting URLs by examining phishing URL properties, including lexical and hosted features (Sadique et al., 2020). They introduced a model for identifying phishing websites and demonstrated its high accuracy through testing on various datasets. The researchers tackled various types of smishing and phishing assaults in their work (Mishra and Soni, 2022). They introduced a URL verification mechanism to authenticate the URLs embedded in SMS messages, and they investigated the utility of the smishing-box method to mitigate the risk associated with downloading files during a smishing attack.

Researchers have proposed a spam SMS detection model named 'SMSAssassin,' which combines the Naive Bayes method and Support Vector Machine (Kamau et al., 2022). The system uses crowdsourcing to maintain a current database, with users contributing by updating spam keywords. Additionally, they developed a content-based SMS filtering system and a user-friendly mobile application for SMS management. The researchers carried out a thorough examination of phishing attempts (Jain and Gupta, 2022). They studied several mobile phishing attempts made by threat actors along with the accompanying defences. The research included a classification of countermeasures against phishing attempts, author strategies, various difficulties associated with identifying phishing assaults, and the development of several phishing datasets. Other researchers have addressed smishing detection techniques, discussing various taxonomies for identifying mobile phone phishing in their study (Do et al., 2022). They also highlighted different phishing strategies, including voice phishing, Bluetooth-based phishing, and mobile app phishing, contributing to user and researcher awareness. Scholars developed a method for detecting phishing endeavours on mobile devices named 'PhisherCop' (Noah et al., 2022).

The prior research highlights a critical limitation in the application of machine learning algorithms when exclusively focused on URL links and hyperlinks. While machine learning is undoubtedly a valuable tool in the domain of smishing detection, it is apparent that a more comprehensive approach is essential to address the diverse and evolving nature of smishing tendencies. Like other cyber threats, smishing attacks use a wide variety of tactics and techniques, which makes them difficult to detect. To effectively combat smishing, other factors besides the URLs, such as the content of the message need to be considered. This will allow us to develop a more robust and versatile smishing detection system that can adapt to the constantly changing landscape of cyberattacks. To address this gap, we formulated the following research question: How can a multi-feature ensemble learning model improve malicious SMS detection by leveraging textual content analysis?

Table 1
The dataset's record count.

Dataset	Smish	Ham	Total
DATASET-I	1127	4844	5971
DATASET-II	747	4827	5574
Total	1874	9671	11 545

3. Material and methods

This study addresses the detection of smishing (SMS phishing) attacks using an ensemble learning approach. By leveraging these models, the study enhances detection accuracy and robustness in identifying malicious SMS messages. To tackle class imbalance, the SMOTE is applied, improving generalization and preventing overfitting. The research employs TF and TF–IDF for feature extraction, allowing for more effective differentiation between smishing and legitimate messages. The dataset was split into 80% for training and 20% for testing using random sampling, a standard practice in machine learning. This ensures balanced representation of both Smish and legitimate (Ham) classes, enabling the model to generalize effectively. The ensemble model outperforms individual classifiers, offering a robust solution for smishing detection, and the publicly available datasets ensure the reproducibility of the results. This section introduces the dataset, the techniques for feature extraction, and the machine learning models incorporated within the experimental framework. A visual representation of the experimental approach is provided in Fig. 2.

3.1. Dataset

In the present study, experiments are carried out to detect smishing SMS using a combination of two distinct datasets. To ensure an adequately expansive dataset for experimentation, both datasets are amalgamated into a unified dataset. Dataset-I denoted as 'SMS PHISHING DATASET',[1] has been procured from Mendeley, while dataset-II, titled 'SMS Smishing Collection Dataset',[2] has been sourced from Kaggle.

For clarity, the Mendeley dataset contains three categories: Smishing, Ham, and Spam. In our study, we have combined the Smishing and Spam categories into a single 'Smish' category to represent all malicious messages, while 'Ham' represents legitimate SMS messages. In the Kaggle dataset, which is divided into Spam and Ham, we interpret the Spam category as 'Smish' to align with our focus on smishing detection. This ensures consistency across both datasets. The distribution of records within each dataset is illustrated in Table 1.

[1] https://data.mendeley.com/datasets/f45bkkt8pr/1
[2] https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset/data

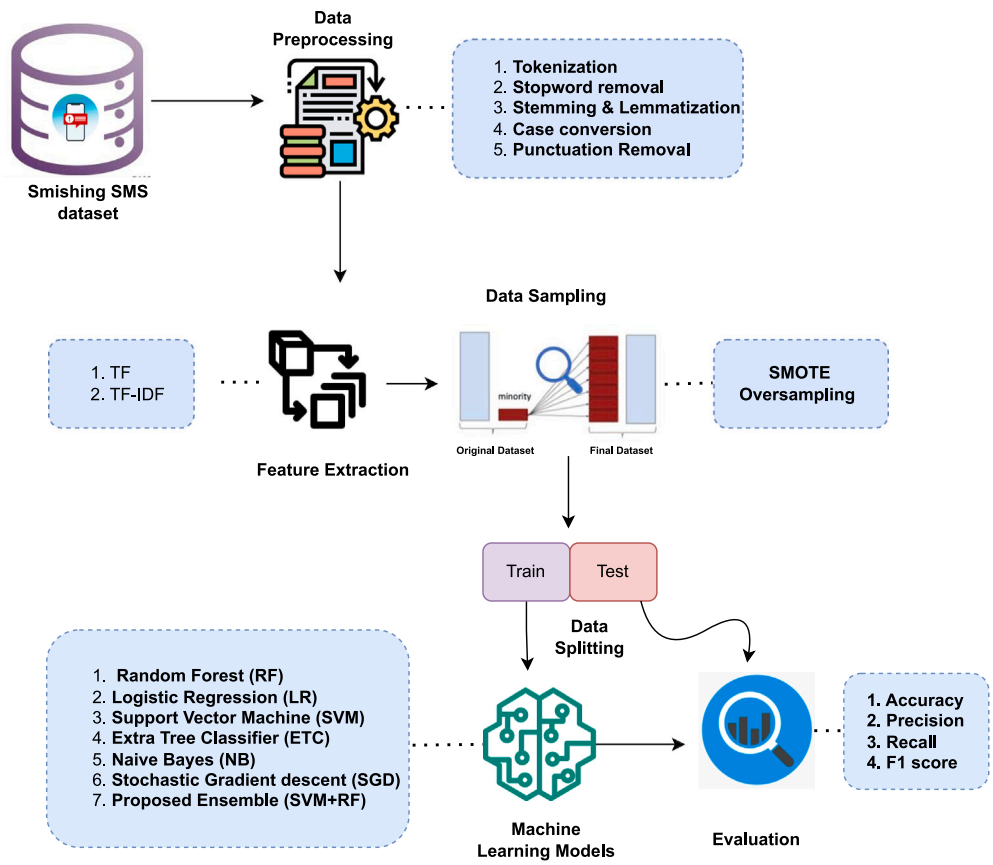

Fig. 2. Architecture of methodologies adopted for machine learning models.

3.1.1. Preprocessing steps

The datasets contain data that is either unstructured or semi-structured, which may not inherently convey meaningful information. The presence of such extraneous input can decelerate the model training process, potentially impeding the model's overall performance. To enable machine learning models to operate with greater efficiency while conserving computational resources, preprocessing becomes imperative. Enhancing the model's capability to deliver reliable predictive outcomes entails text preparation, encompassing tasks such as tokenization, converting text to lowercase, removing stopwords, and eliminating numbers.

In this study, various preprocessing techniques are applied, encompassing actions like the removal of numbers, elimination of stopwords and punctuation, conversion of text to lowercase, stemming, and lemmatization. The elimination of raw data serves to reduce feature complexity since text data can often contain superfluous content that might adversely influence model performance. The natural language toolkit (NLTK) package in Python is employed for the preprocessing operations.

3.2. Feature extraction

To facilitate the training of classifiers, textual content needs to be translated into numerical representations. This conversion is essential for preserving the information contained in the text. Several techniques, such as Term Frequency (TF) and Term Frequency–Inverse Document Frequency (TF–IDF), can be employed to assign numerical values to each word, enabling this transformation.

3.2.1. TF

In natural language processing and information retrieval, term frequency (TF) is a numerical representation that is used to measure how frequently a term or word appears in a document or text corpus. It calculates a word's frequency of occurrence based on the total number of words in a text. TF is frequently computed mathematically using the formula:

$$TF = \frac{\text{(Number of times the term appears in the document)}}{\text{(Total number of terms in the document)}} \tag{1}$$

The derived value signifies the term's significance and occurrence within the document, serving as a fundamental element in various methodologies employed for activities such as text analysis, information retrieval, and machine learning applications like document categorization and the ranking of search engine results

3.2.2. TF-IDF

The TF–IDF method assists in the detection of significant and unique terms that can effectively distinguish documents within a given collection. It finds applications in text classification, information retrieval, and keyword extraction, among various other natural language processing tasks. By incorporating TF–IDF values, text data representation is enhanced, as it considers not only the frequency of words within specific documents but also the rarity of terms across the entire corpus.

$$W_{i,j} = [1 + log(tf_{i,j})] \times [log(\frac{N}{df_i})] \tag{2}$$

In this context, N stands for the overall number of documents, $TF_{i,j}$ represents the term's frequency within a particular document, and $D_{f,t}$ pertains to the document that encompasses the text t.

3.3. Oversampling

This study employed the Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002) which is a useful tool for machine

learning and data preparation. It is mostly used to correct class imbalance in classification. Traditional machine learning algorithms frequently fail to build accurate and trustworthy models when confronted with unbalanced datasets, when one class considerably outnumbers the other(s). This is due to their bias towards the majority class, which has more data to learn from. Using a minority class instance as a starting point, it finds its k nearest neighbours in the feature space, where 'k' is a user-defined parameter. Then, by combining the characteristics of the selected instance with those of its neighbours, synthetic samples are produced. Each synthetic sample is created by randomly selecting one of the k neighbours and adjusting the feature differences between the neighbour and the selected instance by a random number between 0 and 1. To create several synthetic samples, this technique is repeated. By adding these recently generated samples to the dataset, the minority class's representation is essentially increased.

3.4. Machine learning models

This section presents supervised machine learning models utilized in the experiments. The models including Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), Stochastic Gradient Decent (SGD), Naive Bayes (NB) and an ensemble of RF and SVM are employed in combination with Tf and TF–IDF to attain results.

3.4.1. RF

Random Forest (RF) is a tree-based adaptable machine learning approach which is used for both classification and regression applications. It functions by aggregating the forecasts from decision trees, every tree is trained using a distinct subset of the data. This lowers the chance of overfitting and increases prediction accuracy. Moreover, RF is resistant to data noise, which makes it a suitable option for bigger datasets. Numerous domains, such as text categorization, natural language processing, and fraud detection (Lombo et al., 2022), use the widely used RF method. It may be used to rate the significance of features in the data and is rather simple to apply and evaluate.

3.4.2. LR

logistic regression (LR) (Das, 2021) is a statistical technique and kind of regression analysis used to model the likelihood of a binary outcome. It is typically applied to text classification issues in machine learning (Shah et al., 2020). The logistic regression model determines the likelihood that an input falls into one of two classes. For binary classification, logistic regression is useful because it uses the logistic or sigmoid function to transform outputs into class probabilities (0 and 1). To compute log odds, the model finds feature coefficients. These log odds are then converted into probabilities. Its simplicity makes it a popular option that may be expanded to include complicated interactions. Additionally, feature coefficients provide information about predictive significance and LR is less likely to overfit.

3.4.3. SVM

Support Vector Machine (SVM) is a supervised machine-learning algorithm for regression and classification problems (Pisner and Schnyer, 2020). Its main use case is in classification tasks, where it performs exceptionally well in identifying the hyperplane that divides several classes in a dataset. To provide a decision boundary that is suitable for classification, SVM seeks to identify a hyperplane that maximizes the margin in the classes. Tasks involving multi-class classification as well as binary classification may be handled with SVM. While SVM can handle numerous classes in multi-class issues using one-vs-all or one-vs-one techniques, it finds a hyperplane that divides the data into two classes in the case of binary classification.

3.4.4. ETC

An ensemble learning method for classification is called the Extra Trees Classifier (ETC) (Geurts et al., 2006). While it provides an additional degree of randomization by randomly selecting features and splitting thresholds, it is comparable to Random Forest. By making the model more unpredictable, overfitting is less likely to occur and the model is more resilient to noisy or dispersed input. To increase efficiency, particularly with high-dimensional datasets, ETC uses bagging and regularization techniques. Its hyperparameters need to be properly adjusted to get optimal performance. Because of its adaptability and ability to handle complicated data, ETC is extensively utilized in many different industries.

3.4.5. NB

A popular option for classification jobs is the probabilistic machine learning technique Naive Bayes (NB) (Webb et al., 2010), which is especially useful in fields like natural language processing and text analysis. The Bayes theorem is utilized to calculate the probability of data points falling into particular categories. Though Naive Bayes is named "naive" because it makes a significant assumption about feature independence, it works well in a variety of real-world scenarios. It has several versions that are specific to different types of data, such as Bernoulli, Gaussian, and Multinomial. When it comes to high-dimensional data situations, like text analysis, Naive Bayes performs exceptionally well and provides valuable insights into the relative relevance of distinct variables. Its uses include sentiment analysis and spam identification (Rodrigues et al., 2022).

3.4.6. SGD

A popular optimization technique in machine learning is stochastic gradient descent (SGD) (Liu et al., 2020), which is especially useful for training different models such as linear models. It is an optimization method that is based on gradient descent and has several benefits, particularly when working with big datasets. SGD is a flexible optimization technique that performs well, especially in online learning settings and with big datasets. It is efficient, flexible, and effective. It has been used in text categorization task (Vidyashree and Rajendra, 2023).

3.5. Proposed ensemble

The approach for classifying malicious SMS is presented in this part, along with an overview of the proposed framework. The proposed framework utilizes an ensemble model called SVM+RF, uniting two distinct machine-learning models for malicious SMS detection. Fig. 3 provides a visual depiction of the framework's architecture. The use of ensemble models has grown in prominence in recent years, resulting in better accuracy and effectiveness in classification outcomes. It is feasible to improve performance beyond the capability of individual models by combining several classifiers. An ensemble learning strategy is used in this work to increase the prediction capabilities of malicious SMS detection. The proposed method includes a voting classifier that combines the RF and SVM models using soft voting criteria. The class with the most votes determines the final prediction.

$$\hat{p} = argmax\{\sum_{i}^{n} SVM_i, \sum_{i}^{n} RF_i\}. \tag{3}$$

The prediction probabilities for each test instance are derived by subjecting it to both the SVM and RF models, represented as $\sum_i^n SVM_i$ and $\sum_i^n RF_i$, respectively. Then, the probabilities corresponding to each test case, resulting from the SVM and RF models, are subjected to the soft voting criterion, as depicted in Fig. 3.

$$SVM + RF = argmax(g(x)) \tag{4}$$

The ensemble model (RF+SVM) determines the final class by selecting the class with the highest average probability and combining

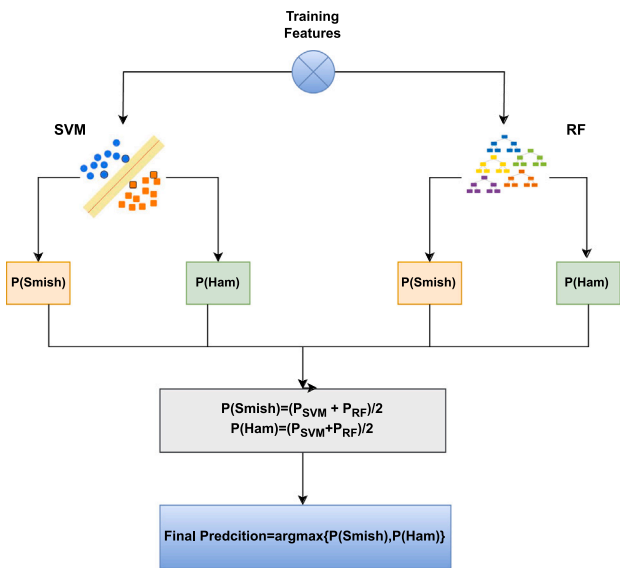

Fig. 3. Architecture of the proposed ensemble model.

the projected probabilities from both classifiers. This paper proposes a system for improving the categorization accuracy of malicious and ham SMS messages by merging two benchmark datasets from publicly available repositories. To improve the quality of the text and make it simpler to handle, the dataset is cleaned first using a series of preprocessing techniques. Then the dataset is balanced by applying SMOTE. The dataset is then divided into 70:30 halves for training and testing. The SVM+RF architecture is used to successively integrate two classifiers that are both well-suited for processing textual input. This, in turn, leads to better categorization accuracy.

3.6. Evaluation metrics

Evaluating the models requires assessing their ability to classify SMS based on commonly recognized criteria. These criteria include precision, recall, and the F1 score. Accuracy provides a high-level view of a model's skill by measuring the fraction of properly categorized cases compared to the full dataset, demonstrating overall correctness.

$$Accuracy = \frac{\text{Number of correctly classified predictions}}{\text{Total predictions}} \tag{5}$$

while in the case of binary classification, accuracy is measured as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

While the terms can be defined as follows:

True positives (TP) are the number of accurate positive forecasts that match the number of actual positive incidents.

False positives (FP) signify the quantity of negative forecasts that are inaccurately identified as positive.

True negatives (TN) are the number of exact negative forecasts that correlate to real-world negative cases.

False negatives (FN) represent the count of incorrect positive forecasts that clash with the factual negative instances.

Precision measures the model's accuracy in classifying positive instances by comparing the fraction of correctly categorized positive occurrences to the total predicted positive occurrences. This statistic emphasizes the model's ability to reduce false positives.

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

Recall, alternatively termed sensitivity, gauges the model's proficiency in accurately recognizing positive instances. This metric is derived by dividing the count of true positives by the complete pool of positive samples. Its relevance becomes particularly accentuated in contexts where the precise detection of positive cases carries paramount significance, exemplified in scenarios such as disease diagnosis

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

The F1 score, a widely adopted metric in binary classification scenarios, leverages the harmonic mean to amalgamate accuracy and recall into a singular figure. This property renders it particularly advantageous when handling imbalanced datasets. It plays a pivotal role in harmonizing the balance between precision and recall, with elevated scores indicative of superior model performance.

$$F1score = 2\frac{precision.recall}{precision + recall} \tag{9}$$

Table 2
Classification result on Imbalanced Dataset.

Feature	Model	Accuracy	Precision	Recall	F1-Score
TF	RF	95.45%	95%	95%	95%
	LR	93.23%	93%	93%	93%
	SVM	95.14%	95%	95%	95%
	ETC	93.62%	94%	94%	94%
	NB	93.54%	92%	92%	92%
	SGD	91.64%	91%	91%	91%
	SVM+RF	96.21%	96%	96%	96%
TF–IDF	RF	95.89%	96%	96%	96%
	LR	94.35%	94%	94%	94%
	SVM	95.75%	95%	95%	95%
	ETC	94.58%	94%	94%	94%
	NB	95.32%	95%	95%	95%
	SGD	93.25%	93%	93%	93%
	SVM+RF	96.39%	96%	96%	96%

4. Results and discussion

This section contains an in-depth examination of the experimentation carried out for this research. The machine learning models used in experiments are RF, LR, SVM, ETC, NB and SGD. These models are trained with hyperparameters that were carefully optimized for the Smishing dataset. Fine-tuning these hyperparameters entailed experimenting with parameter value ranges to get optimal performance.

4.1. Performance of machine learning models on imbalanced dataset

This section presents the experiments carried out on the original imbalanced dataset in this research, followed by a detailed analysis of the obtained results. The research focuses on the results of tests targeted at identifying malicious SMS messages using various feature engineering approaches. Using frequency-based approaches such as TF and TF–IDF, we systematically analyse the performance of several supervised machine learning models. These models, which include RF, LR, SVM, ETC, NB, SGD, and SVM+RF, are tested for accuracy, precision, recall, and the F1 score. The findings show that the efficiency of each model is dependent on the feature extraction strategy used.

Table 2 compares the performance of several machine learning models on the unbalanced dataset using TF and TF–IDF. The results show that when using the TF approach, RF surpasses the others, reaching a 95.45% accuracy and a 95% precision, recall, and F1 Score. Similarly, SVM achieves equivalent results, with 95% precision, recall, and F1 score for Smishing SMS categorization. Furthermore, ensemble SVM+RF performs admirably, with an accuracy of 96.21%, precision, recall, and F1 score of 96%.

Interestingly, using TF–IDF does not appear to have resulted in any substantial improvement in classifier performance. Among the models examined, SVM+RF with TF–IDF produced the best results, with an accuracy score of 96.39% and precision, recall, and F1 scores all at 96%. Meanwhile, LR and ETC produced comparable results, with 94% value of precision, recall, and F1 scores for identifying Smishing SMS.

Table 3
Classification result on the Balanced Dataset.

Feature	Model	Accuracy	Precision	Recall	F1-Score
TF	RF	98.26%	98%	97%	98%
	LR	96.05%	96%	96%	96%
	SVM	97.79%	97%	97%	97%
	ETC	96.32%	96%	96%	96%
	NB	96.39%	96%	96%	96%
	SGD	95.45%	96%	96%	96%
	SVM+RF	98.39%	98%	98%	98
TF–IDF	RF	99.10%	98%	98%	98%
	LR	95.10%	96%	96%	96%
	SVM	98.56%	98%	98%	98%
	ETC	96.89%	97%	97%	97%
	NB	97.10%	97%	97%	97%
	SGD	96.32%	96%	96%	96%
	SVM+RF	99.58%	99%	99%	99%

Table 4
Classification result of other Ensemble models Using TF–IDF on the Balanced Dataset.

Model	Accuracy	Precision	Recall	F1-Score
RF+SGD	98.24%	98%	97%	98%
LR+SGD	96.58%	96%	95%	96%
SVM+ETC	97.33%	98%	97%	98%

4.2. Performance of machine learning models on the balanced dataset

Table 3 presents the performance of classifiers on the balanced dataset utilizing TF and TF–IDF. The dataset is balanced by applying SMOTE. Results reveal that the performance of classifiers is significantly improved when applied to the balanced dataset. RF outperformed the other individual models for malicious SMS classification using the TF, with an accuracy of 98.26% and precision, recall and F1 scores of 98%,97% and 98% respectively. SVM achieved the second highest results, with an accuracy of 97.79% and precision, recall, and F1 scores of 97%. The ensemble model SVM+RF performed better than individual models with an accuracy of 98.39% and precision, recall, and F1 scores of 98%.

All classifiers have shown improved results using TF–IDF on the balanced dataset for malicious SMS detection. It is worth noting that RF and SVM have shown robust results on the balanced dataset using TF–IDF as individual models with 99.10% and 98.56% accuracy respectively. SVM+RF have outperformed other models with 99.58% accuracy, 99% score of precision, recall and F1-score.

4.3. Results of ensemble models

In this study, three sets of ensemble earning models are also evaluated on the balanced dataset using TF–IDF for comparative purposes. These ensemble models include RF+SGD, LR+SGD, and SVM+ETC. Within the scientific community, these models have gained popularity in text categorization applications. Among these three models, RF+ETC has shown the highest results. The ensemble recurrent model SVM+RF, on the other hand, stands out by giving remarkable performance. The SVM+RF outperforms on a variety of assessment parameters, including accuracy, precision, recall, and F1 score. These findings support the usefulness of the SVM+RF model proposed in this paper, which benefits from an ensemble structure that combines the capabilities of SVM and RF (see Table 4).

4.4. Discussion

On the large-sized dataset created by integrating two benchmark datasets for malicious SMS detection, the performance of machine learning models and ensemble models is compared. To handle the class imbalance problem, SMOTE is applied. The influence of two feature engineering strategies, TF and TF–IDF, is also tested in conjunction with the models on imbalanced and balanced datasets. Fig. 4 shows a comparison of classifiers on the imbalanced dataset and Fig. 5 shows the comparison of classifiers on the balanced dataset based on accuracy, precision, recall, and F1 score. The results clearly show that ensemble models outperform their machine-learning counterparts separately. However, RF+SVM exceeds other models with a 99.58% accuracy and a 99% value of precision, recall, and F1 score.

The ensemble SVM+RF model's effectiveness in detecting malicious SMS can be ascribed to its ability to efficiently use the large size of data generated by combining two datasets. The strong performance of this model across a range of assessment measures and its capacity to recognize subtle patterns make it an important tool for text categorization and anomaly identification. The study emphasizes how crucial it is to use learning algorithms with an appropriate feature extraction technique and data resources to succeed at the difficult task of malicious SMS detection.

The proposed malicious SMS detection approach, which uses ensemble learning and SMOTE, may be easily implemented into mobile cybersecurity systems in various ways. First, the model may be directly incorporated into end-user mobile security applications. These applications may execute the malicious SMS detection model locally on the smartphone, offering real-time security by analysing incoming SMS messages and alerting them to probable phishing attempts. Because of the model's lightweight design may be implemented efficiently on mobile devices with minimal computational overhead. Mobile service providers can deploy a malicious SMS detection system at the network level, which scans SMS messages before transmitting them to consumers. This would enable the early identification and filtering of malicious messages, providing proactive protection against malicious SMS attacks. Finally, the model may be integrated with current mobile device management platforms, especially in business environments. This would allow enterprises to secure their workers' mobile devices against malicious SMS attacks by monitoring and filtering SMS conversations via a centralized security solution.

4.4.1. Comparison of performance with cutting-edge models

A complete comparison performance study is done against various cutting-edge techniques in the field of malicious SMS detection to assess the significance of the SVM+RF model described in this research. In this regard, two recent research papers have been chosen that have performed classification on malicious SMS. In a study conducted by Ulfath et al. (2022), they used SVM to detect malicious SMS attacks on a dataset of 5572 SMS messages, getting an accuracy score of 98.39%. The TF–IDF method was used for feature extraction in their approach. Agrawal et al. (2023) used a successful technique in their research, employing a hybrid machine learning model incorporating RF, NB, and ETC. Their study used a dataset of 5169 SMS messages, yielding a great accuracy score of 96.9%.

Table 5 presents a thorough comparison of the previously described research efforts in combination with the results produced from the proposed model provided in this work. The investigation demonstrates that when combined with SMOTE, the SVM+RF ensemble model beats previous techniques, obtaining a stunning accuracy score of 99.58%. This discovery highlights the suggested model's higher performance, particularly when dealing with a large volume of SMS texts. The findings highlight the effectiveness of ensemble models such as SVM+RF in solving complex and dynamic security concerns. As the threat landscape develops, the adaptability and accuracy of models are important in proactively tackling emerging threats. This study emphasizes the significance of adopting cutting-edge methodologies and data resources to successfully protect against malicious SMS attacks in the digital era.

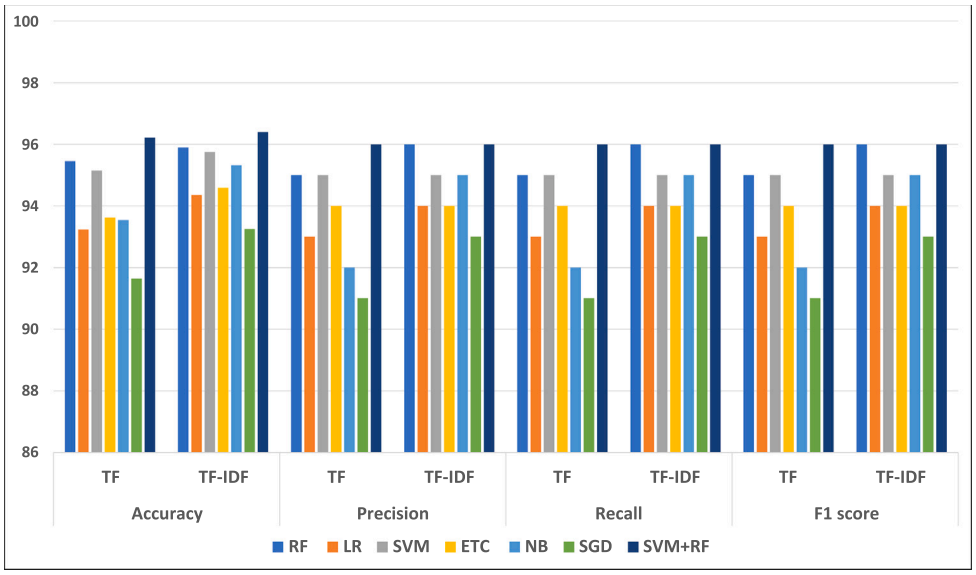

Fig. 4. Performance comparison of models on imbalanced dataset.

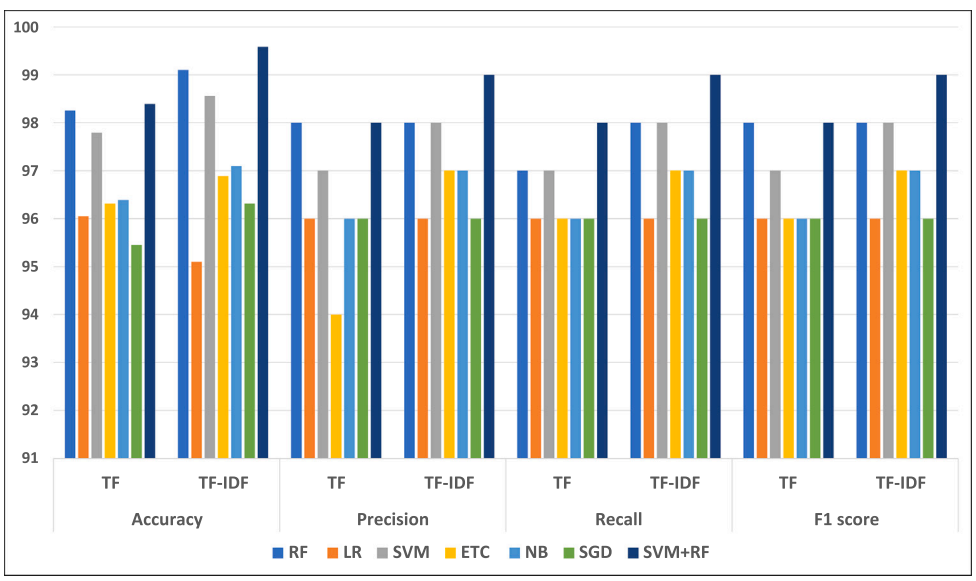

Fig. 5. Performance comparison of models on the balanced dataset.

Table 5
Comparative analysis using cutting-edge models from the literature.

Reference	Dataset Count	Model	Accuracy
Ulfath et al. (2022)	5572 SMS	SVM	98.39%
Agrawal et al. (2023)	5169 SMS	RF+NB+ETC	96.9%
Proposed	11545 SMS	SVM+RF with SMOTE	99.58%

5. Conclusion and future work

The increasing prevalence of smartphones and their constant internet connectivity make individuals susceptible to the growing threat of phishing attempts. Phishers employ deceptive tactics, particularly through text messages, to manipulate individuals into revealing sensitive information or engaging in compromising activities.

This study focuses on addressing the emerging threat of malicious SMS attacks targeting smartphones, which combines smishing attempts and spam SMS to deceive users. While previous attempts have been made to detect malicious SMS, there remains room for improvement in classification results. To tackle this challenge, the research leverages ensemble learning, specifically a fusion of SVM and RF called SVM+RF. To create a substantial dataset, two benchmark datasets are merged, and the issue of class imbalance is mitigated using SMOTE.

This research emphasizes the distinct advantages of ensemble learning in detecting malicious SMS, offering a comprehensive comparison with conventional machine learning models. The results underscore the exceptional performance of this architecture, achieving an accuracy score of 99.58% in the identification of malicious SMS. The proposed technology automates SMS categorization by detecting SMS categories on its own, removing the need for human participation. This study's direction may be further by using deep learning approaches to improve result accuracy, which would be beneficial in the SMS classification process.

CRediT authorship contribution statement

Hongsheng Xu: Investigation, Funding acquisition, Data curation, Conceptualization. **Akeel Qadir:** Resources, Project administration, Investigation, Formal analysis. **Saima Sadiq:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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